

作品集

Portfolio

射频电路设计 · 天线系统 · SIPI仿真 · 电磁仿真 · AI驱动优化 · 全链路设计验证

2027届应届硕士 | GPA 3.64 | 年级第一

3段实习经历 | 5+项目经验 | SCI论文+专利+竞赛

应聘人：袁宏伟

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1.68GHz 相控阵雷达系统 — 落地能力

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项目概述

设计并实现完整的相控阵雷达天线系统，包括天线单元仿真优化、阵列综合、

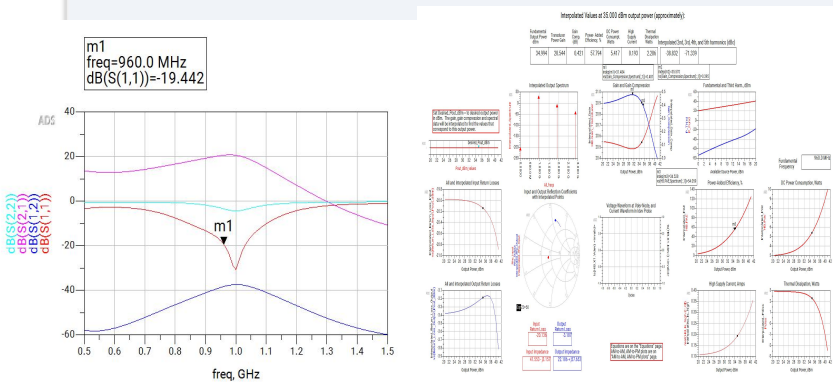
波束调控、功率放大器及功分器。覆盖从电磁仿真到硬件实测全流程。

技术栈：HFSS · ADS · 矢量网络分析仪 · 测试

PA功率放大器

LDMOS设计
S参数仿真优化

非线性分析



八单元方向可调相控阵天线系统设计与实现:



相控阵天线射频系统



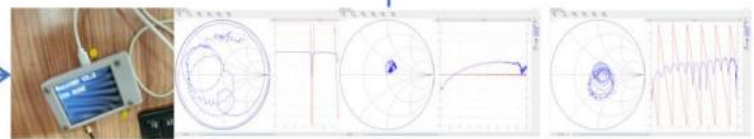
天线S参数测试
以及馈线的相位补偿



8端口移相器的功率, 相位赋值



波束方向测试

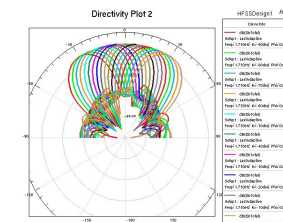
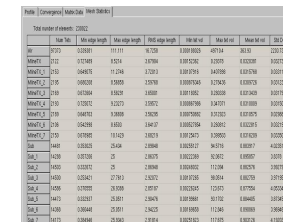
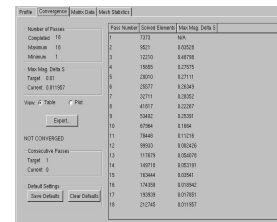
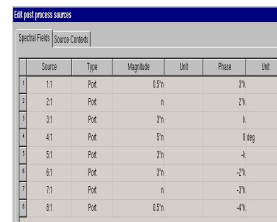
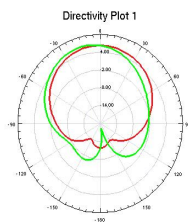
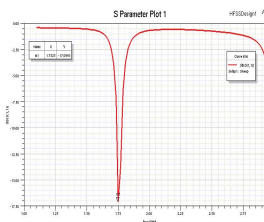
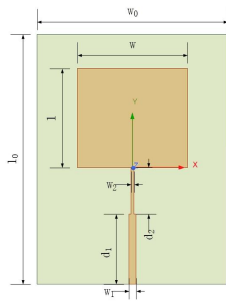


1.1

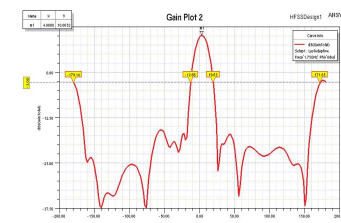
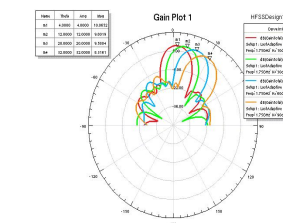
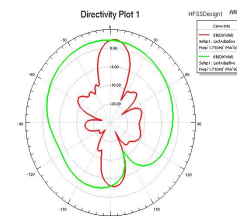
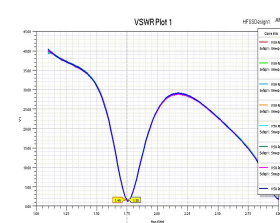
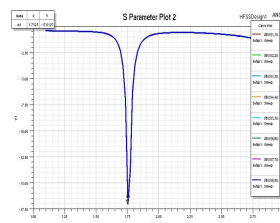
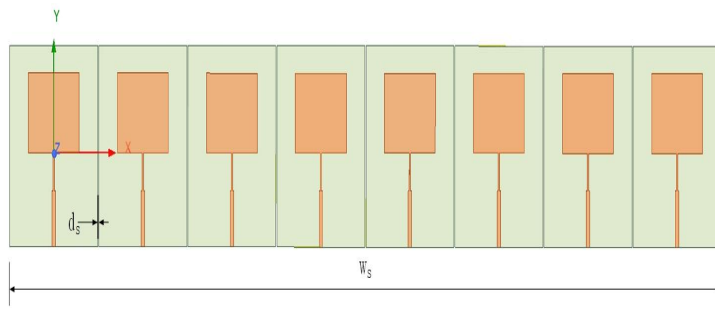
相控阵天线 - 电磁仿真 (HFSS)

Phased Array Antenna - EM Simulation Results

天线模型



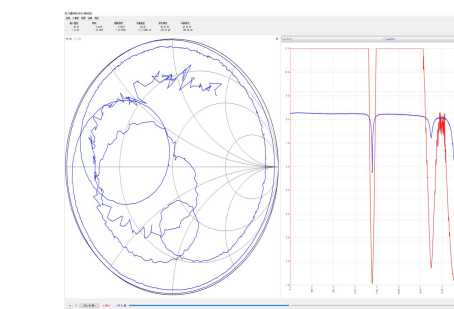
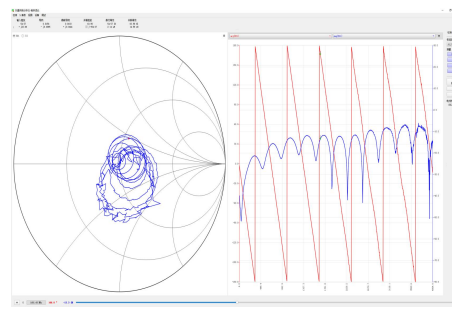
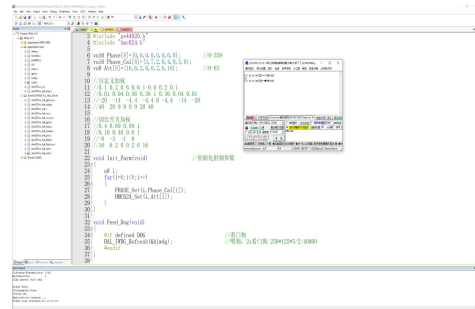
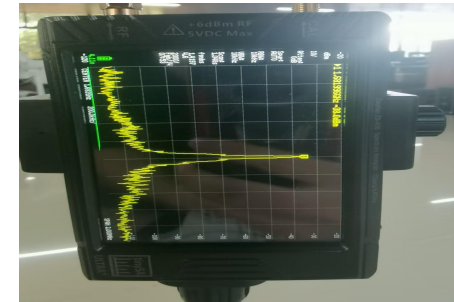
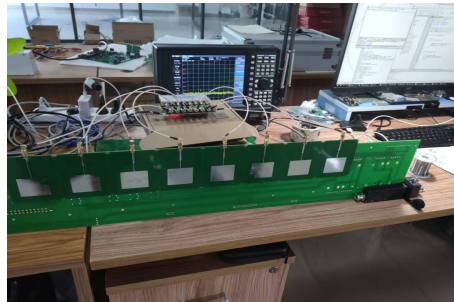
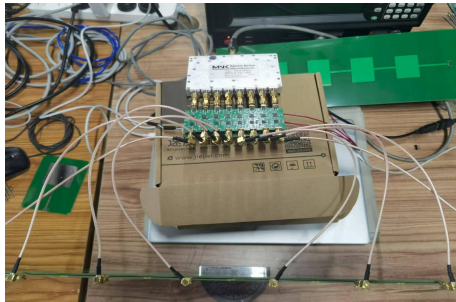
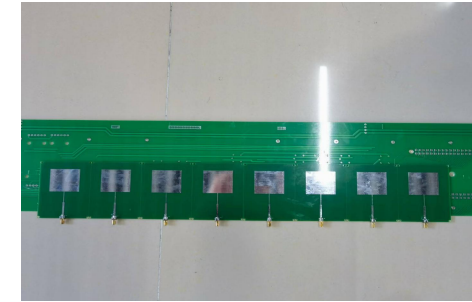
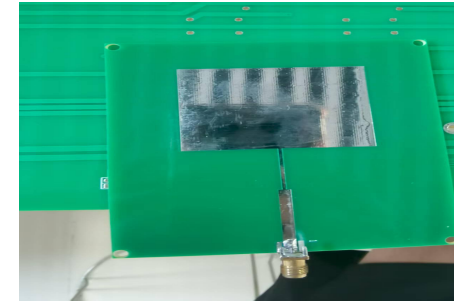
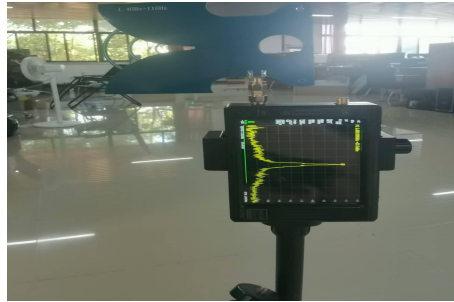
阵列天线



1.2

相控阵天线 - 硬件实测

Phased Array Antenna - Hardware Measurement



2

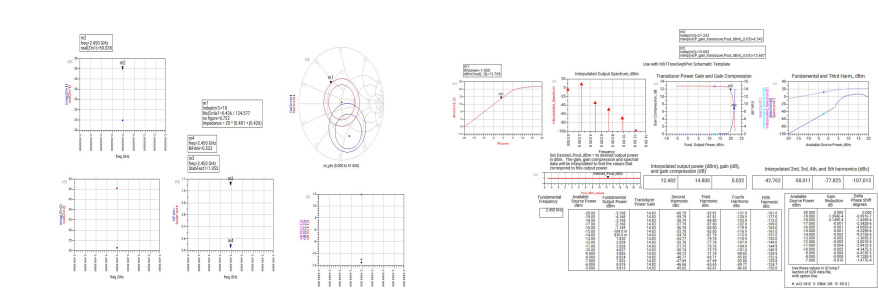
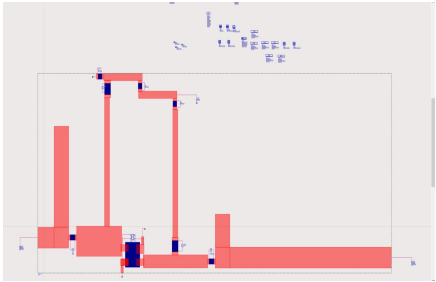
2.45GHz 低功耗 LNA 设计-联调能力

2.45GHz Low Power Low Noise Amplifier Design

项目概述

设计一款工作在2.45GHz ISM频段的低功耗低噪声放大器，
满足无线通信接收前端对噪声系数和功耗的严格要求。

设计工具：Keysight ADS

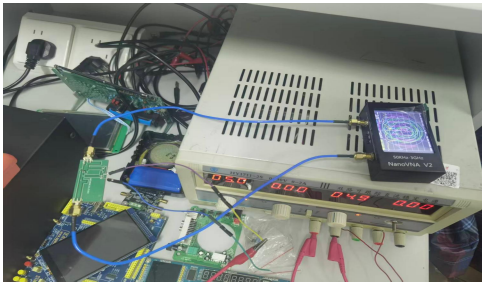
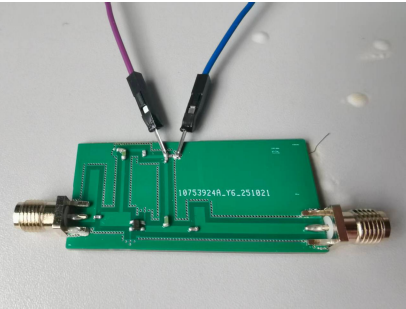
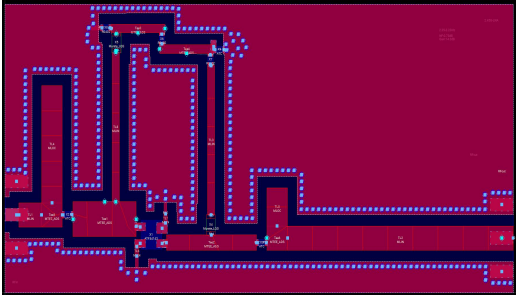


NF<1 dB
噪声系数

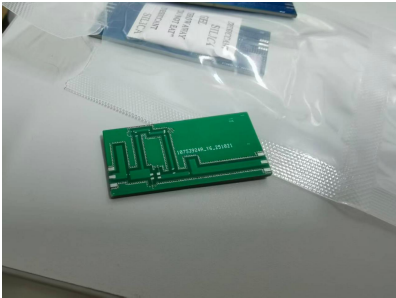
Gain≈15 dB
增益

Pdc<10 mW
功耗

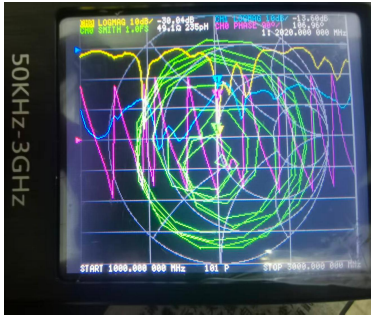
K>1
稳定性



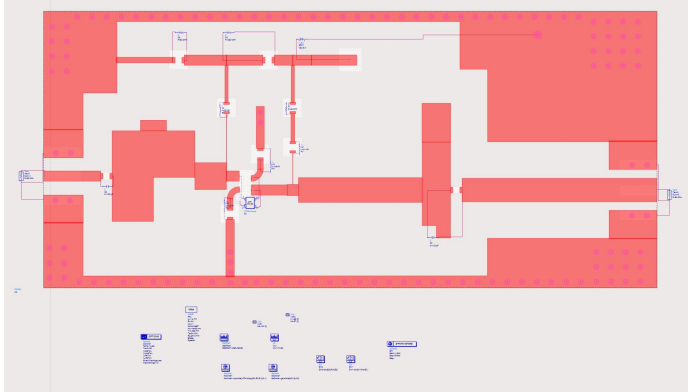
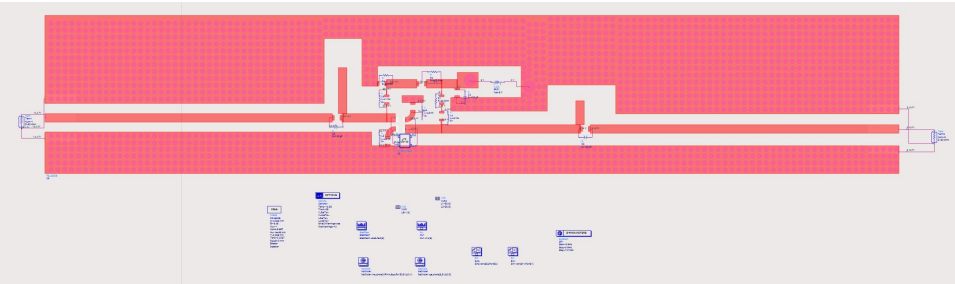
手动焊接
0402RLC，
ATF5414chip



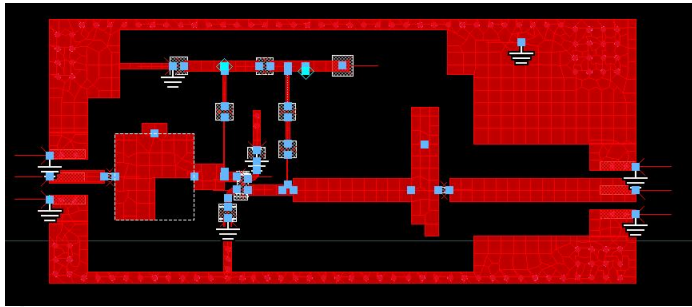
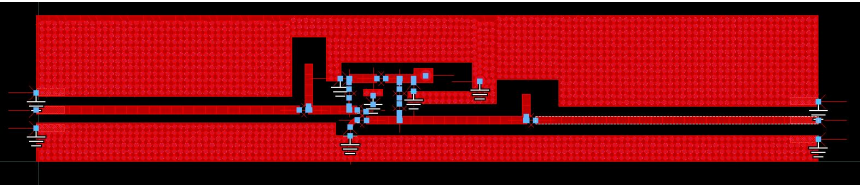
焊接器件对应
锡焊位置放入
加热台，调至
245左右
，出现锡膏融
化，即可关闭
加热台，完成
焊接



2.1 2.45GHz-LNA（RO4350B与FR-4切换基板之后的设计）



联调

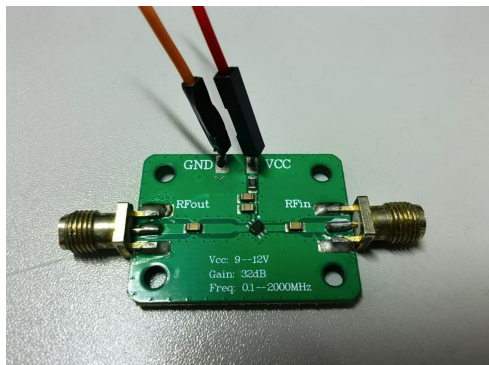


Substrate



性能指标	S21/dB	S11/dB	S22/dB	NF/dB	OP1/dBm	二次谐波抑制/dBc	稳定性
RO4350B	13.53	-13.18	-13.51	0.54	21.25	-61.95	>1.05
FR-4	13.57	-28.7	-17.54	0.55	17.53	-79.37	>1.05

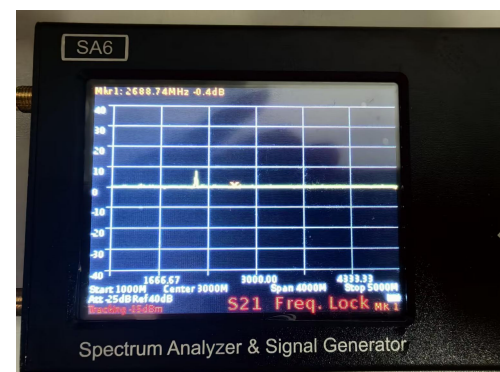
2.2 LNA硬件测试



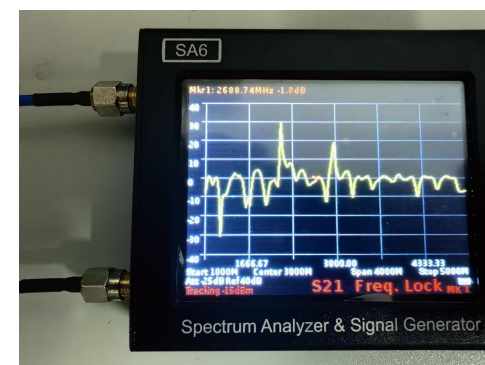
LNA实物



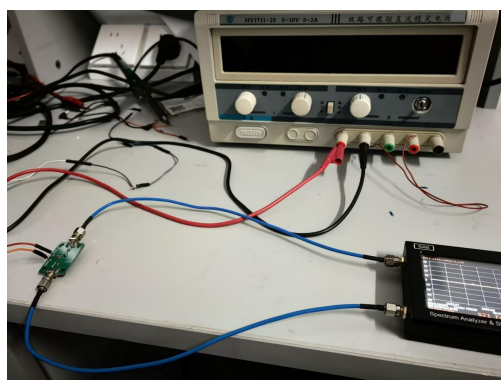
VNA测试



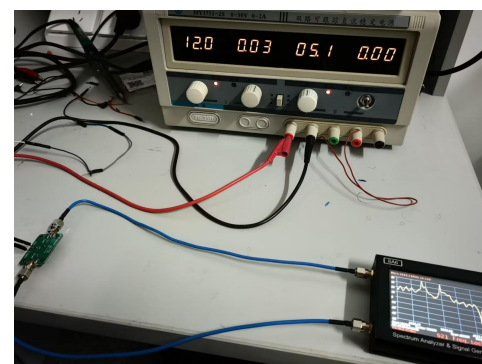
频谱仪SA6归一化



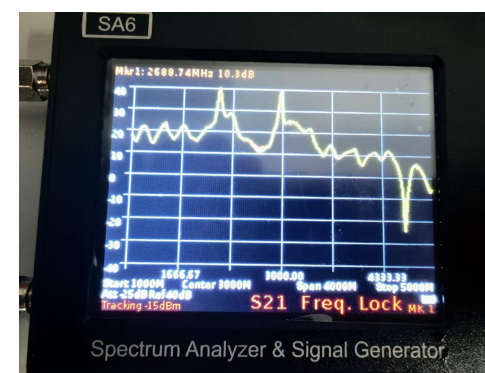
接入线缆的 S_{21}



接入LNA未加电源的 S_{21}



接入LNA加电源的 S_{21}

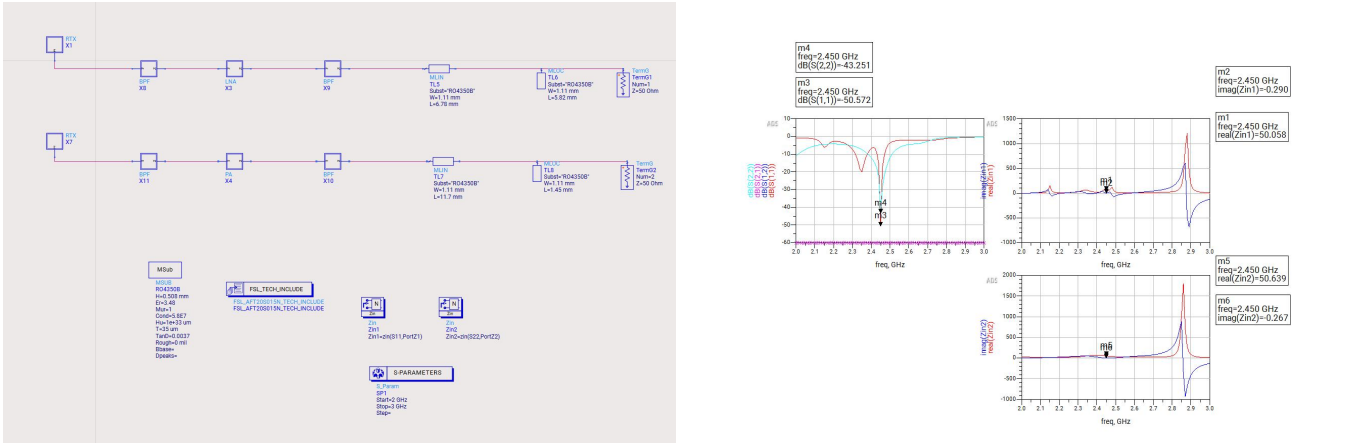


项目概述

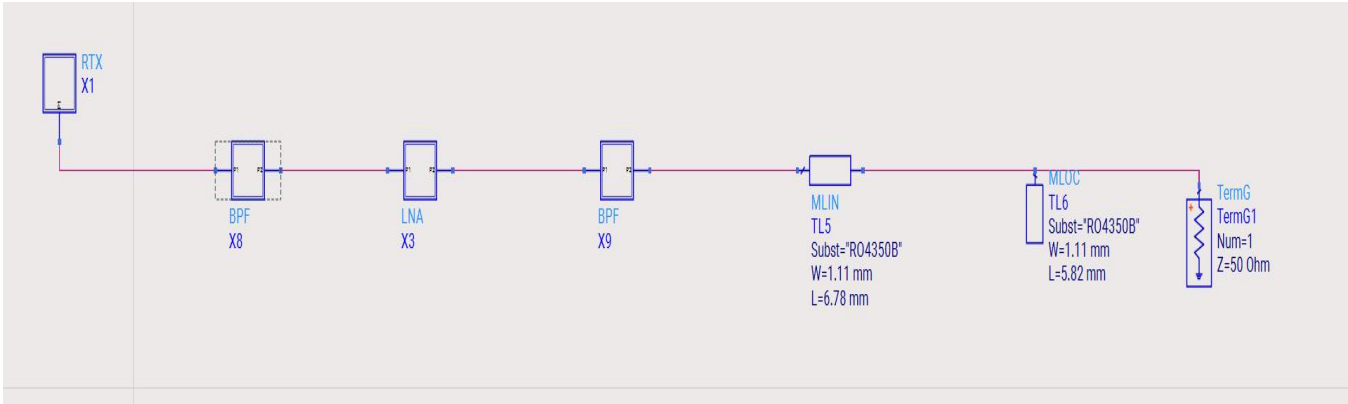
设计完整的2.45GHz射频收发链路，包含发射链路和接收链路，完成系统链路预算、模块指标分配及整体性能仿真验证。

设计工具：Keysight ADS

系统链路仿真



接收机链路仿真结果



链路预算

增益/噪声/线性度

频率规划

中频选择与镜像抑制

模块分解

TX-BPF-PA-BPF
RX-BPF-LNA-BPF

版图设计

PCB布局与阻抗控制

4 AI-Agent 驱动的射频设计系统

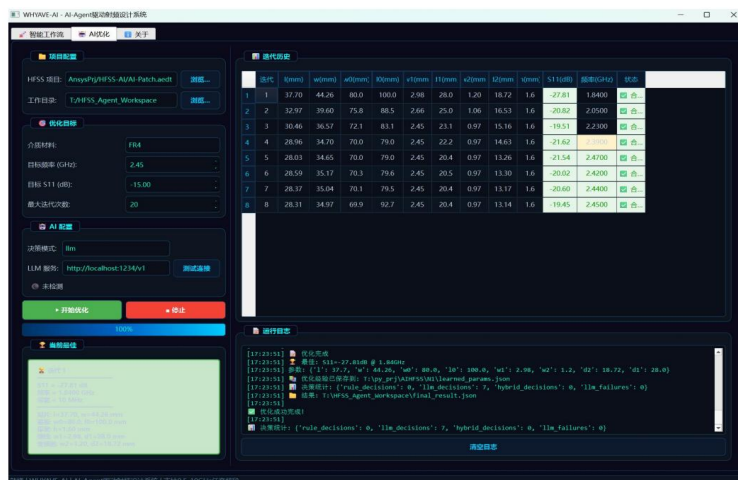
AI-Agent Driven RF Design System

项目概述

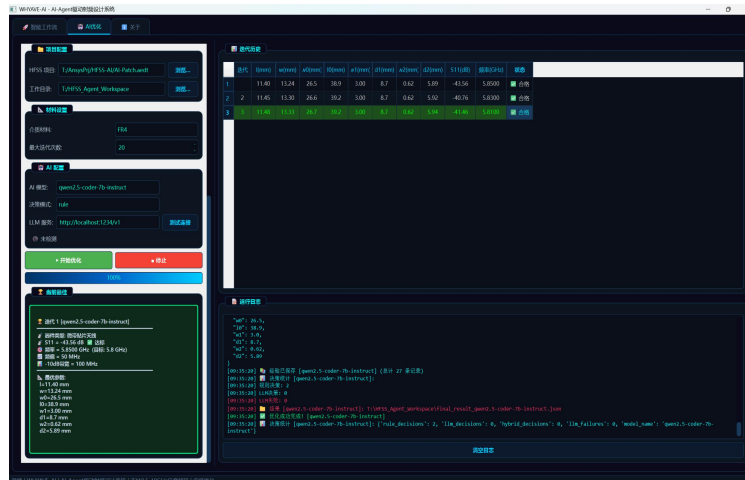
开发基于大语言模型(LLM) Agent的射频电路智能设计系统，实现从需求输入到

电路方案输出的全自动化设计流程。支持2.45GHz和5.8GHz双频段射频前端电路的智能设计与优化。

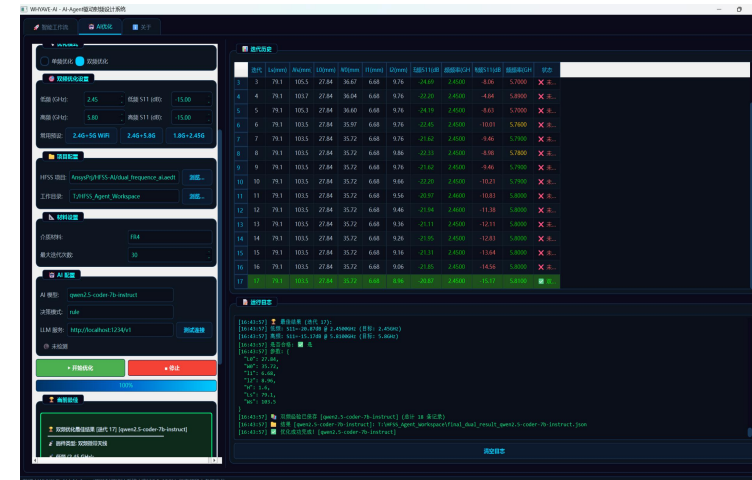
2.45GHz 设计结果



5.8GHz 设计结果



双频段联合设计



LLM Agent

大模型驱动
智能设计决策
自动迭代优化

知识库

射频电路设计
知识图谱
经验规则库

仿真集成

ADS/HFSS/CST
API自动调用
结果自动分析

单/双频段

2.45G+5.8G
频段覆盖
架构可扩展

产品化

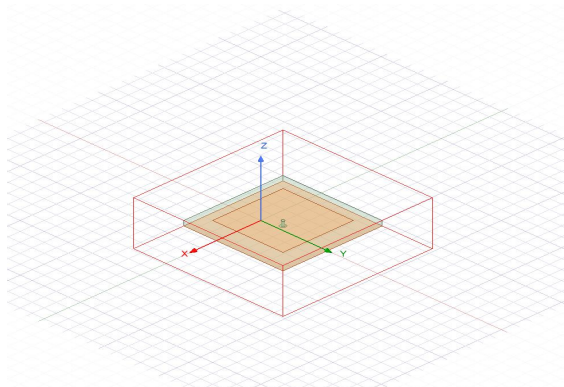
完整产品手册
标准化流程
可复制部署

项目概述

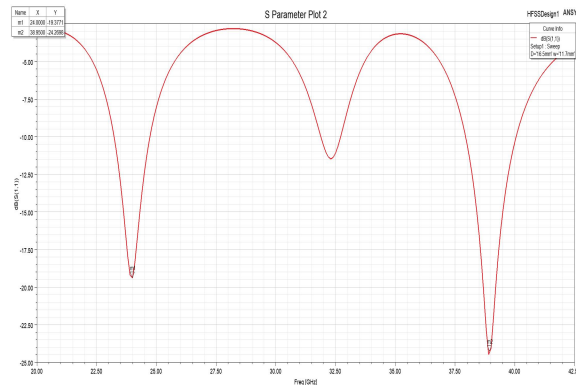
设计24GHz和77GHz两个毫米波频段的相控阵天线，适用于无人机、雷达和5G通信应用。

完成天线单元设计、阵列布局优化、波束扫描仿真及关键性能参数评估。

24GHz 相控阵天线

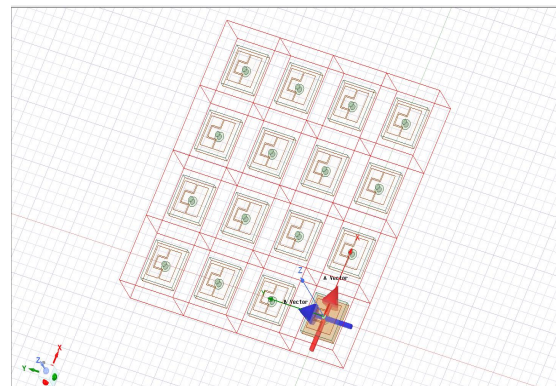


24GHz天线模型

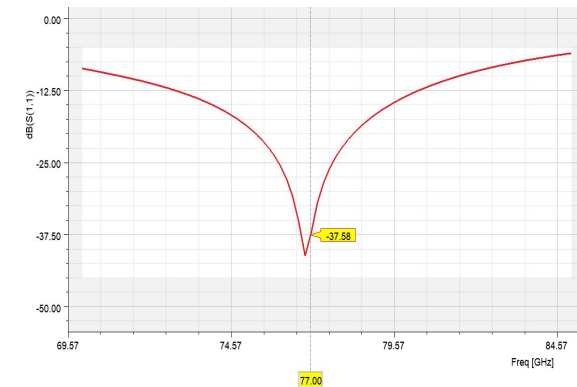


24GHz S参数仿真

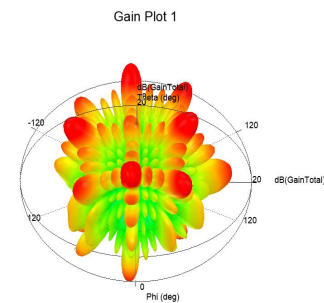
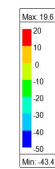
77GHz 相控阵天线



77GHz天线模型



77GHz S参数仿真



ANSYS

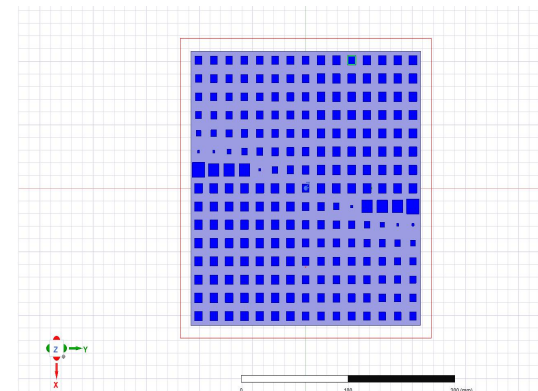
项目概述（技术栈：HFSS · MATLAB · 超表面单元设计 · 相位调控）

（1）突破孔径极限：OAM涡旋电磁波感知

传统高分辨感知依赖大阵列天线，小型无人机物理孔径先天不足。

（2）OAM超表面：解锁太赫兹机载成像

OAM超表面以亚波长结构对电磁波进行像素级相位操控，将复杂光学系统压缩为一片可共形贴附的智能蒙皮。这让小型无人机首次具备太赫兹成像能力，为种子内部缺陷、果实隐性损伤的无损检测打开关键硬件入口。



超表面单元

亚波长超表面
单元实现360
相位覆盖

OAM模式

产生 $l = \pm 1, \pm 2$
多种OAM模式
模式纯度>85%

宽带设计

宽带超表面设计
覆盖完整工作频段
带内性能稳定

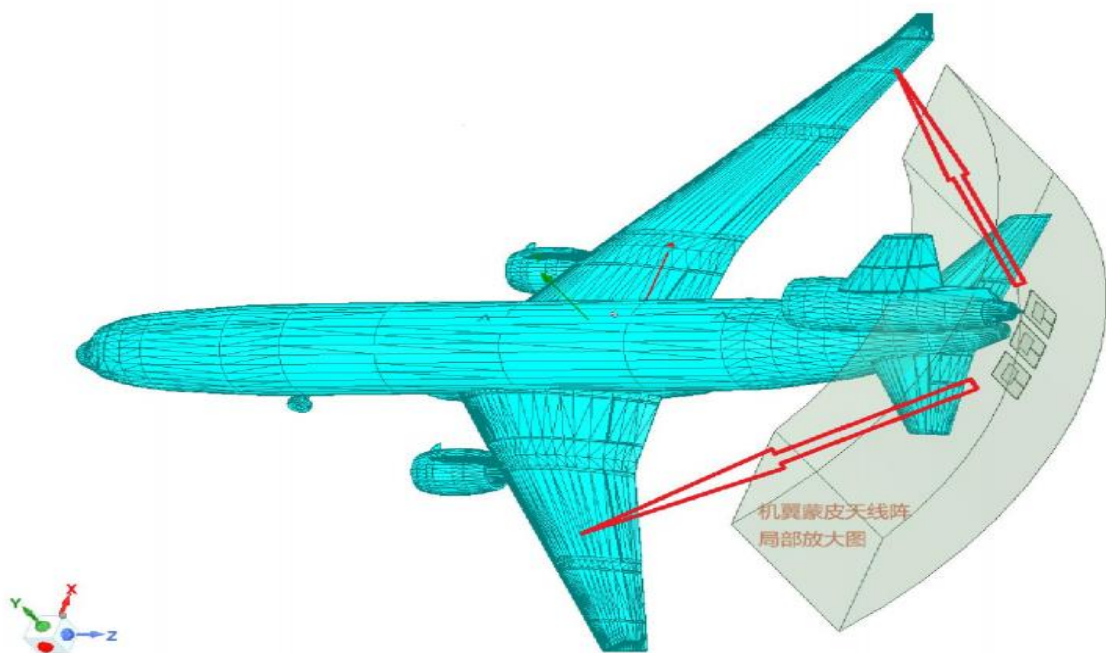
应用前景

6G通信·雷达成像
量子信息·大容量
无线传输

项目概述

研究弯曲与温度对机翼蒙皮共形微带阵列天线 S_{11} 参数的耦合影响，建立了多物理场耦合分析模型。

该研究成果已发表于SCI期刊，是共形天线领域的重要理论贡献。



研究成果

SCI期刊论文撰写

建立了弯曲-温度多物理场耦合分析模型，揭示了

机翼蒙皮形变对天线 S_{11} 参数的影响规律。

理论创新

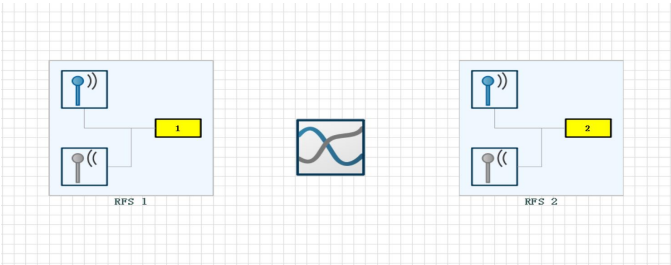
- 首次系统研究弯曲+温度双因素耦合效应
- 提出共形天线S参数偏移补偿方法
- 为机载共形天线设计提供理论指导

项目概述

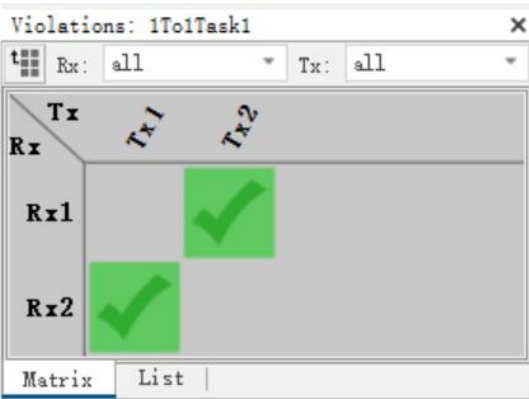
开发自动化工具实现通信系统串扰数据的批量导出、清洗、分析和可视化后处理，大幅提升测试数据处理效率。

整机布局

技术栈：CST · MATLAB · 自动化脚本 · 数据可视化



数据处理流程



Component Selection Tree

Navigation Tree

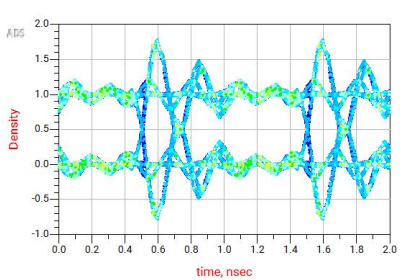
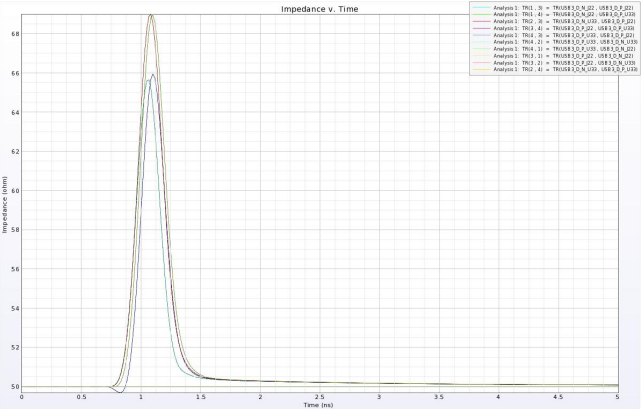
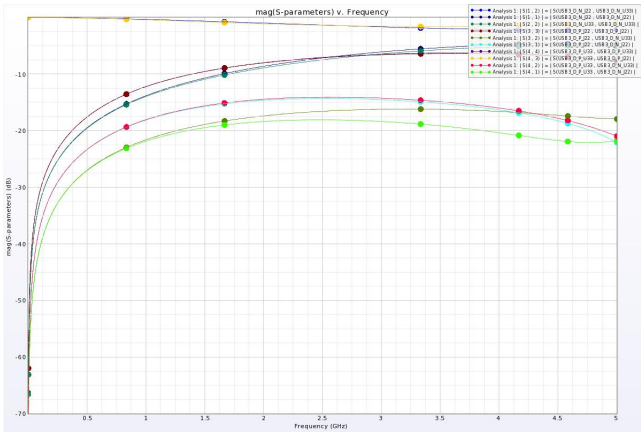
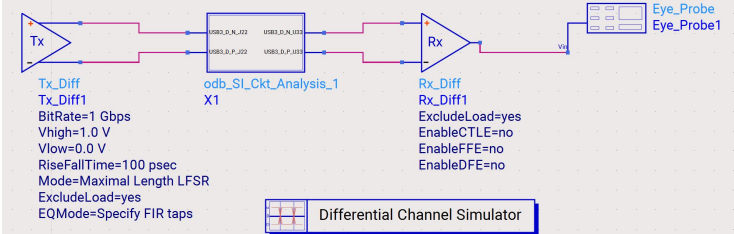
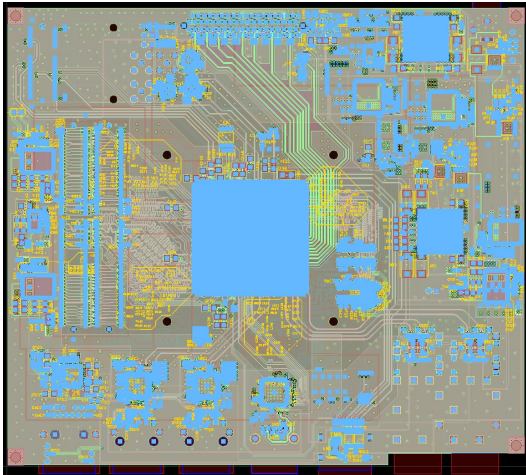
Violations: 1To1Task1

Rx: Rx1

Tx: Tx2

Violation	Frequency	Type	Rx	Channel	Tx	Channel
-78.27 dB	1.800 GHz	in-band	Rx1	Channel 5	Tx2	Channel 4
-105.87 dB	2.000 GHz	in-band	Rx1	Channel 6	Tx2	broadband
-88.88 dB	2.001 GHz	out-of-band	Rx1	Channel 1	Tx2	Channel 6
-88.88 dB	2.001 GHz	out-of-band	Rx1	Channel 2	Tx2	Channel 6
-88.88 dB	2.001 GHz	out-of-band	Rx1	Channel 3	Tx2	Channel 6
-88.88 dB	2.001 GHz	out-of-band	Rx1	Channel 4	Tx2	Channel 6
-88.88 dB	2.001 GHz	out-of-band	Rx1	Channel 5	Tx2	Channel 6
-78.27 dB	2.001 GHz	in-band	Rx1	Channel 6	Tx2	Channel 4
-76.37 dB	2.001 GHz	in-band	Rx1	Channel 6	Tx2	Channel 5
-68.88 dB	2.001 GHz	in-band	Rx1	Channel 6	Tx2	Channel 6
-98.27 dB	2.600 GHz	out-of-band	Rx1	Channel 1	Tx2	Channel 4
-98.27 dB	2.600 GHz	out-of-band	Rx1	Channel 2	Tx2	Channel 4
-98.27 dB	2.600 GHz	out-of-band	Rx1	Channel 3	Tx2	Channel 4
-98.27 dB	2.600 GHz	out-of-band	Rx1	Channel 4	Tx2	Channel 4
-96.37 dB	2.800 GHz	out-of-band	Rx1	Channel 1	Tx2	Channel 5
-96.37 dB	2.800 GHz	out-of-band	Rx1	Channel 2	Tx2	Channel 5
-96.37 dB	2.800 GHz	out-of-band	Rx1	Channel 3	Tx2	Channel 5
-96.37 dB	2.800 GHz	out-of-band	Rx1	Channel 4	Tx2	Channel 5
-96.37 dB	2.800 GHz	out-of-band	Rx1	Channel 5	Tx2	Channel 5

案例二



index	permute(Height)	permute(Width)
0.000	0.722	7.400E-10

time	Density
index=0.000	
605.0 psec	-0.668
1.605 nsec	-0.668
610.0 psec	-0.616
1.610 nsec	-0.616
610.0 psec	-0.601
1.610 nsec	-0.601
620.0 psec	-0.563
1.620 nsec	-0.563
620.0 psec	-0.508
1.620 nsec	-0.508
1.620 nsec	-0.507
630.0 psec	-0.483
1.630 nsec	-0.483
850.0 psec	-0.463
1.850 nsec	-0.463
850.0 psec	-0.457
1.850 nsec	-0.457
850.0 psec	-0.452
1.850 nsec	-0.452
850.0 psec	-0.445
1.850 nsec	-0.445
850.0 psec	-0.434
1.850 nsec	-0.434
850.0 psec	-0.432
1.850 nsec	-0.432
850.0 psec	-0.431
1.850 nsec	-0.431
850.0 psec	-0.430
1.850 nsec	-0.429

基于Keysight ADS平台，对DDR高速总线进行完整SI仿真分析。通过TDR阻抗测试、S参数提取及眼图仿真，验证了PCB走线的信号质量。仿真结果显示：特性阻抗稳定控制在 $50\Omega \pm 1\%$ 范围内，插入损耗在5GHz频点仅为-2.19dB，眼图张开充分、抖动小、噪声容限充足。整体信号完整性表现优异，满足高速接口设计要求，验证了PCB叠层结构与走线布局的合理性。仿真初期遇到的S参数无源性报错已确认为求解器收敛性问题，与设计质量无关。

10 科研成果 - 两篇SCI

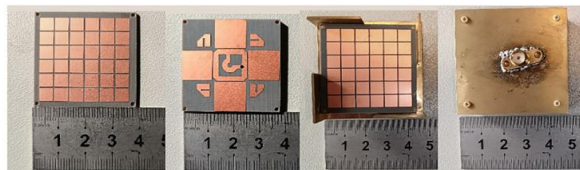


FIGURE 10 | Antenna prototype.

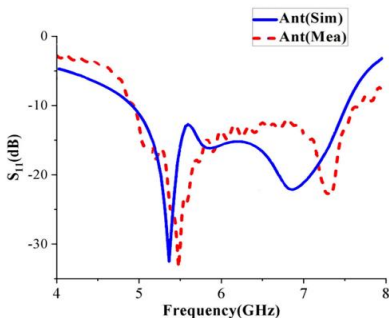


FIGURE 11 | Simulated and measured S_{11} values.

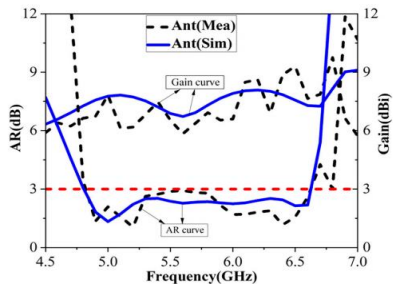


FIGURE 12 | Simulated and measured broadside Gain and the axial ratio (AR).

TABLE 2 | Comparison of the proposed antenna to previously reported antennas.

Reference	Volume (λ^3)	MS	No. of layers	Impedance bandwidth (%)	AR bandwidth (%)	Peak Gain (dBi)
[13], 2017	$1.45 \times 1.452 \times 0.028$	No	1	15.9	11.8	12.5
[14], 2017	$0.92 \times 0.92 \times 0.028$	No	1	19.51	12.9	9.8
[4], 2020	$1.06 \times 1.06 \times 0.03$	No	1	23	18.9	10.2
[15], 2021	$1.02 \times 1.02 \times 0.028$	No	1	25.8	20.6	8
[24], 2024	$0.66 \times 0.66 \times 0.065$	Yes	3	19.7	19.9	4.86
[25], 2025	$1.58 \times 1.58 \times 0.02$	No	1	2.25	0.25	8.8
[26], 2025	$0.97 \times 0.97 \times 0.057$	Yes	2	16.42	19.74	8.4
Our work	$0.82 \times 0.82 \times 0.20$	Yes	2	43.4	31.4	9.1

Abbreviations: AR, axial ratio; MS, metasurface.

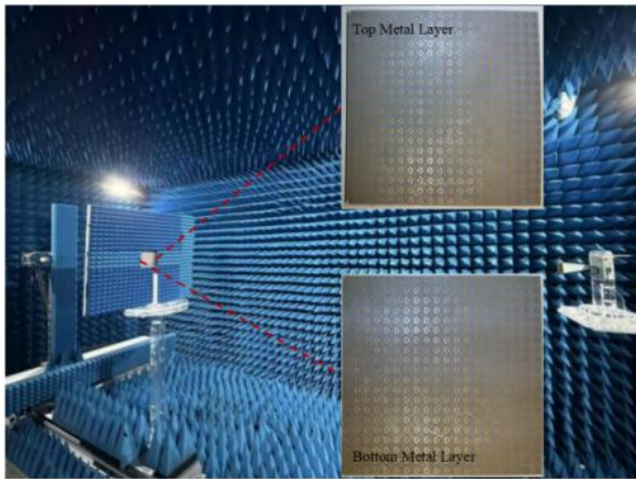


Fig. 9: Experimental setup and measurement environment, and enlarged view of the metasurface array

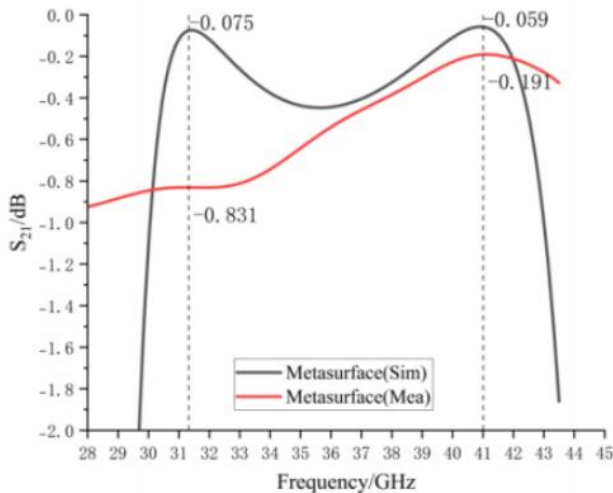


Fig. 10: Comparison of S_{21} Simulation and Measurement

TABLE III: Comparison of this FSS design with reported designs

Reference	Operating Frequency	Layer Number	Relative Bandwidth	S_{21} / Power Transmittance
[26]	26.2 GHz	2	15.7% (24.1–28.2 GHz)	-0.89 dB/81.5%
[29]	0.6 THz	3	18.75% (0.58–0.7 THz)	$\approx 90\%$
[30]	9 GHz, 15.5 GHz	2	4.44%/6.45%	-0.76 dB/84%; -0.6 dB/87%
[31]	6.5 GHz, 10.5 GHz	4	10.8%/11.4%	-0.41 dB/91%; -0.14 dB/97%
Proposed	31.33 GHz, 41.01 GHz	2	33.74% (30.3–42.6 GHz)	-0.831 dB/82.57%; -0.191 dB/95.69%

解决的是“宽带圆极化天线的小型化与性能增强”问题，适用于5G/6G室内通信、卫星通信等场景。

解决的是“高频段双频FSS的高透射、低剖面与结构简化”问题，适用于5G毫米波透镜天线、波束成形系统等。

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技能与工具总结

Skills & Tools Overview

电磁仿真	Ansys HFSS	CST Microwave	Keysight ADS	Cadence SpectreRF	Altium Designer
测试测量	矢量网络分析仪	频谱分析仪	信号源	暗室测试	探针台测试
编程开发	Python	MATLAB	C/QT	VBA	Shell脚本
设计领域	天线设计	射频电路	功放设计	毫米波	EMC/EMI
AI/自动化	LLM Agent	机器学习	自动化脚本	数据处理	可视化

感谢观看

THANK YOU

期待与您的交流！

射频电路设计 · 天线系统 · 电磁仿真 · AI驱动优化 · 全链路验证